Daniel Hardesty Lewis

Recipient
Naresh Gambirapuram
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To whom it may concern,

Scale, performance, and reliability define my work experience at the Texas Advanced Computer Center. Leading a team of myself alone, I designed and scaled a terrain aggregation pipeline toprovide Texas's first seamless elevation raster product, one of the principal products of a $40M disaster resiliency initiative and a project which led to millions of dollars in contracts with the State of Texas and the US Army Corps of Engineers. This necessitated maxing out 800,000 nodes on one of the top-5 largest computers in the world, authorizing and spending close to $1M in public funds on this project. Recognizing the immense commercial value of this work from its earliest stages, I created a commercialization and licensing strategy and pitched this strategy to UT's director of IP, and gained approval with no substantive changes. I then led a team to design and implement a sales portal to provide this data to energy, civil engineering, and architecture firms across the state of Texas:

summitgeospatial.com

How did I do it?

using the full suite of skills this job requires:
- ingesting gigabytes of shapefile input
- outputting gigs of geojson and gdb files
- querying in PostGIS at the largest possible scales
- against metadatabase tables of millions of complex polygons
- and sometimes, for performance, manipulating these polygons at the WKT level
- ingesting terabytes of img, dem, & tif and outputting TBs of gtiff files
- executing perhaps some of the largest scale GDAL/OGR pipelines ever run
- including Python-based spatial libraries where needed
- and validating my results in Q throughout

misc skills that may be of value for this work:
- parallel and distributed cloud computing at the highest level
- Dockerized workflow orchestration on the supercomputing equivalent of Airflow
- mentoring of grad, PhD students, and professionals on their own data engineering projects
- deep understanding of commercialization and licensing strategies
- and a keen sense of where the business value of geospatial products might be

If you are interested, feel free to reach out: (713) 371-7875.

Sincerely,

Daniel Hardesty Lewis

Attached: curriculum vitæ

Daniel Hardesty Lewis
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Education

2017 Dec. **B.S., Mathematics**, *University of Texas*, Austin, United States

2017 Dec. **Certificate, Scientific Computation**, *University of Texas*, Austin, United States

Experience

Professional

2023 Nov. – Present **Founder**, *Summit Geospatial*, Austin, United States
Detailed accomplishments:
- Led a team that developed web sales portal of the highest quality seamless terrain data in Texas

2021 Aug. – 2023 Aug. **Engineering scientist associate**, *Texas Advanced Computing Center*, Austin, United States
Detailed accomplishments:
- Principal developer of the highest quality seamless terrain data in Texas
- Lead of a principal project of a $40M disaster resiliency initiative
- Validated data from top State hydrologists and eminent UT engineers
- Developed the licensing & commercialization strategy for this data, secured approval of these strategy from director of IP without substantive changes
- Achieved a 14% click-through rate (CTR) with key audiences
- Helped secure multi-million dollar contracts with the State of Texas and the US Army Corps of Engineers
- Teach undergrads, grads, post-docs, professional researchers from UT Austin, University of Cambridge, & Texas A&M - Galveston

2018 Feb. – 2021 July **Research engineering / scientist associate I**, *Texas Advanced Computing Center*, Austin, United States
Detailed accomplishments:
- Contributed research in integrated modelling and water resource management in Texas and abroad
- Collaborated to aggregate data of Texas's water resources for integrated hydrology modellers

Teaching

2020 Aug. – 2020 Dec. **Co-instructor**, *University of Texas*, Austin, United States
Scientific computation Consulted with students for their homework and class projects

2018 June – 2018 Aug. **Teaching assistant**, *Texas Advanced Computing Center*, Austin, United States
Machine learning for the geosciences Recommended high performance optimisations for Petrobras's code

2018 Feb. – 2018 May **Co-instructor**, *University of Texas*, Austin, United States
Data informatics and intelligent systems for the geosciences Designed and instructed the technical half of this graduate course: intermediate programming techniques, Linux, and introductory high-performance computing

Peer-reviewed articles

2021 July Gil, Y., D. Garijo, D. Khider, C. A. Knoblock, M. Osorio, H. Vargas, M. Pham, J. Pujara, B. Shbita, B. Vu, Y. Chiang, D. Feldman, Y. Lin, H. Song, V. Kumar, A. Khandelwal, M. Steinbach, K. Tayal, S. Xu, S. A. Pierce, L. Pearson, *D. Hardesty Lewis*, E. Deelman, R. F. da Silva, R. Mayani, A. R. Kemanian, Y. Shi, L. Leonard, S. .D. Peckham, M. Stoica, K. M. Cobourn, Z. Zhang, C. Duffy, L. Shu. "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". The ACM Transactions on Interactive Intelligent Systems 11(2):11.

Technology skills

Languages Python, Bash shell, Fortran, Perl

Operating Systems GNU/Linux, macOS, Windows

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Virtualization	Docker, Singularity	Databases	PostgreSQL
Scientific models	MODFLOW, HAND, HEC-RAS	Libraries	GDAL/OGR, GEOS, NumPy, GeoPandas, RasterIO, Matplotlib, Plotly, Dash
Cloud	AWS, Azure	Source code management	Git, GitLab, SVN

Review committees

2023 Mar. – 2023 Apr. Frontera Fellows applications
Helped TACC determine the most qualified doctoral fellows' projects on one of the largest supercomputers in the world

2022 Aug. TACC's Frontera Pathways & Large-Scale Community Partnerships applications
Helped approved/deny millions of dollars in computing resources to multimillion-dollar projects

Students mentored

2021 Aug. – Present Khushboo Agarwal
MS Energy and Earth Resources, the University of Texas at Austin (expected 2023)

2020 Aug. – Present Mark Wang
PhD Civil Engineering, the University of Texas at Austin (expected 2025)
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2022)

2019 Aug. – 2021 May Robert Schomp
MS Environmental and Water Resources Engineering, the University of Texas at Austin (2021)

2019 Aug. – 2020 May Alec Carruthers
MS Environmental Engineering Technology / Environmental Technology, the University of Texas at Austin (2020)

Academic projects

2022 June – 2023 Aug. Advanced compound flood hazard assessment science
Partner with US Army Corps of Engineers to statistically model flood hazards in Texas's coastal region

2021 Oct. – 2023 Aug. Texas Disaster Information System (TDIS)
Provide analytical web-based spatial data system to support more resilient decision-making at the state level

2020 Dec. – 2023 Aug. Museum of South Texas History Sunday Speaker Series
Provide Dash-enabled website to facilitate geolocation of archival imagery by museum visitors

2020 Nov. – 2023 Aug. Networks for Hazard Preparedness and Response
Overlay new, finely tuned flood maps over specially designed building and social vulnerability maps

2020 Sep. – 2023 Aug. Real-time flood inundation mapping to improve community resilience
Provide computationally efficient, high-resolution flood inundation maps to emergency response personnel

2020 Jan. – 2023 Aug. Estimating Inundation Extent and Depth from National Water Model Outputs and High Resolution Topographic Data
Improving the accuracy of flood inundation estimation products 100-fold, from 10m to 1m resolution

Languages

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French Professional working

Spanish Native

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